



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

MATHEMATICS

9709/03

Paper 3 Pure Mathematics 3 (P3)

October/November 2008

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 75.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 4 printed pages.



UNIVERSITY of CAMBRIDGE
International Examinations

- 1 Solve the equation

$$\ln(x + 2) = 2 + \ln x,$$

giving your answer correct to 3 decimal places. [3]

- 2 Expand $(1 + x)\sqrt{1 - 2x}$ in ascending powers of x , up to and including the term in x^2 , simplifying the coefficients. [4]

- 3 The curve $y = \frac{e^x}{\cos x}$, for $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$, has one stationary point. Find the x -coordinate of this point. [5]

- 4 The parametric equations of a curve are

$$x = a(2\theta - \sin 2\theta), \quad y = a(1 - \cos 2\theta).$$

Show that $\frac{dy}{dx} = \cot \theta$. [5]

- 5 The polynomial $4x^3 - 4x^2 + 3x + a$, where a is a constant, is denoted by $p(x)$. It is given that $p(x)$ is divisible by $2x^2 - 3x + 3$.

(i) Find the value of a . [3]

(ii) When a has this value, solve the inequality $p(x) < 0$, justifying your answer. [3]

- 6 (i) Express $5 \sin x + 12 \cos x$ in the form $R \sin(x + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$, giving the value of α correct to 2 decimal places. [3]

(ii) Hence solve the equation

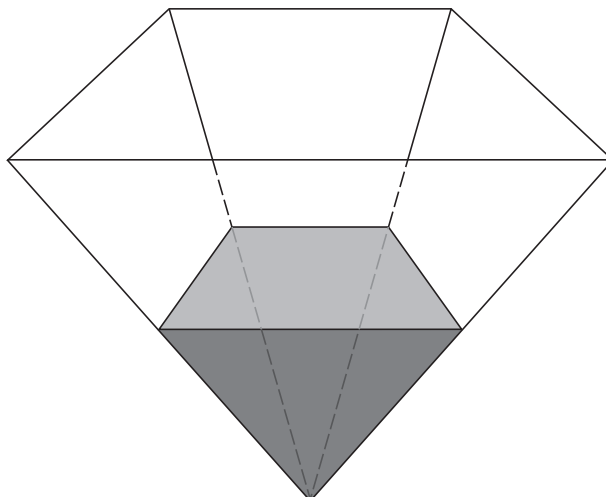
$$5 \sin 2\theta + 12 \cos 2\theta = 11,$$

giving all solutions in the interval $0^\circ < \theta < 180^\circ$. [5]

- 7 Two planes have equations $2x - y - 3z = 7$ and $x + 2y + 2z = 0$.

(i) Find the acute angle between the planes. [4]

(ii) Find a vector equation for their line of intersection. [6]



An underground storage tank is being filled with liquid as shown in the diagram. Initially the tank is empty. At time t hours after filling begins, the volume of liquid is $V \text{ m}^3$ and the depth of liquid is $h \text{ m}$. It is given that $V = \frac{4}{3}h^3$.

The liquid is poured in at a rate of 20 m^3 per hour, but owing to leakage, liquid is lost at a rate proportional to h^2 . When $h = 1$, $\frac{dh}{dt} = 4.95$.

(i) Show that h satisfies the differential equation

$$\frac{dh}{dt} = \frac{5}{h^2} - \frac{1}{20}. \quad [4]$$

(ii) Verify that $\frac{20h^2}{100 - h^2} \equiv -20 + \frac{2000}{(10 - h)(10 + h)}$. [1]

(iii) Hence solve the differential equation in part (i), obtaining an expression for t in terms of h . [5]

9 The constant a is such that $\int_0^a x e^{\frac{1}{2}x} dx = 6$.

(i) Show that a satisfies the equation

$$x = 2 + e^{-\frac{1}{2}x}. \quad [5]$$

(ii) By sketching a suitable pair of graphs, show that this equation has only one root. [2]

(iii) Verify by calculation that this root lies between 2 and 2.5. [2]

(iv) Use an iterative formula based on the equation in part (i) to calculate the value of a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

10 The complex number w is given by $w = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

- (i) Find the modulus and argument of w . [2]
- (ii) The complex number z has modulus R and argument θ , where $-\frac{1}{3}\pi < \theta < \frac{1}{3}\pi$. State the modulus and argument of wz and the modulus and argument of $\frac{z}{w}$. [4]
- (iii) Hence explain why, in an Argand diagram, the points representing z , wz and $\frac{z}{w}$ are the vertices of an equilateral triangle. [2]
- (iv) In an Argand diagram, the vertices of an equilateral triangle lie on a circle with centre at the origin. One of the vertices represents the complex number $4 + 2i$. Find the complex numbers represented by the other two vertices. Give your answers in the form $x + iy$, where x and y are real and exact. [4]